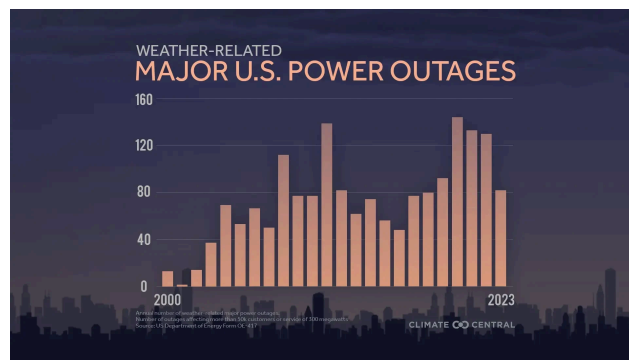


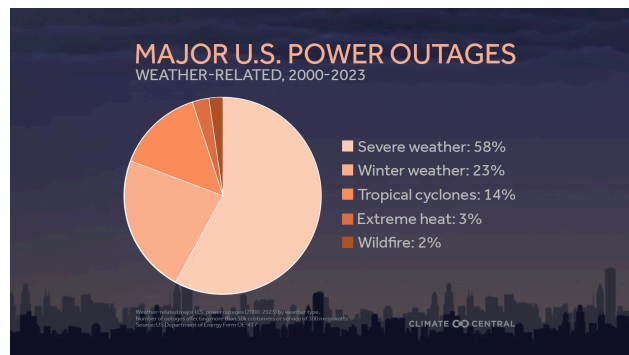
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Weather-related Power Outages Rising



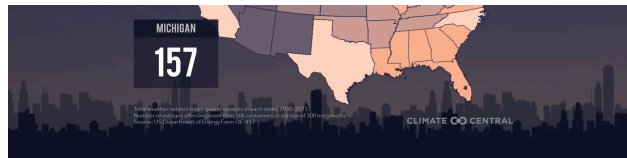
Click the downloadable graphic: Weather Power Outages 2000 to 2023



Click the downloadable graphic: Outages by Weather Type

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Click the downloadable graphic: Weather Outages by State

KEY CONCEPTS

Of all major U.S. power outages reported from 2000 to 2023, 80% (1,755) were due to weather.

Most weather-related outages were caused by severe weather (58%), winter storms (23%), and tropical cyclones including hurricanes (14%).

The states with the most reported weather-related power outages (2000-2023) were Texas (210), Michigan (157), California (145), North Carolina (111), and Ohio (88).

The Southeast (360), South (352), Northeast (350), and Ohio Valley (301) experienced the most weather-related outages from 2000 to 2023.

Download the data

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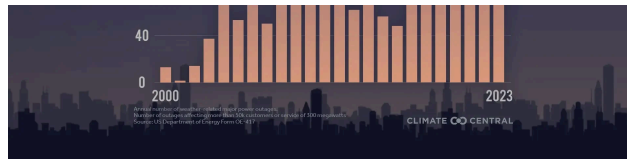
many types of extreme weather are becoming more frequent or intense because of human-caused climate change. These events put stress on aging energy infrastructure and are among the leading causes of major power outages in the U.S.

The nation's electrical grid wasn't built for the present-day climate. Electricity is mostly transmitted and distributed through above-ground transformers, transmission wires, and utility poles that are exposed to extreme weather such as high winds, heavy rain, ice, lightning, and extreme heat. Even in areas where power lines are buried, flooding can lead to loss of power.

Power outages affect millions of people and cost billions of dollars annually. Outages can disrupt access to clean water, food, and critical healthcare. They also have cascading effects on communications networks and transportation.

The impacts of power outages and lengthy restoration times can disproportionately burden people of color, as during the February 2021 Texas cold outbreak; the aftermath of Hurricane Maria in Puerto Rico; and power restoration in Florida after Hurricane Irma.

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Click the downloadable graphic: [Weather Power Outages 2000 to 2023](#)

Weather-related power outages

Climate Central analyzed U.S. power outage data between 2000 and 2023, as [reported by utility companies](#). Major outages are events that affect at least 50,000 customers (homes or businesses) or interrupt service of 300 megawatts or more.

Of all major U.S. power outages reported from 2000 to 2023, 80% (1,755) were due to weather-related events.

Weather-related power outages are on the rise. The U.S. experienced about two times more weather-related outages during the last 10 years (2014-2023) than during the first 10 years analyzed (2000-2009).

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Click the downloadable graphic: Outages by Weather Type

Outages by weather type

Weather-related outages were sorted into five categories: severe weather, winter weather, tropical cyclones (including hurricanes), extreme heat, and wildfire. During the 24-year period analyzed:

Severe weather — such as high winds, rain, and thunderstorms — caused 58% of weather-related outages (1,011 events).

Winter weather — including snow, ice, and freezing rain — accounted for 23% of weather-related outages (398 events).

Tropical cyclones (including hurricanes) caused 14% of weather-related outages (249 events). Although tropical storms accounted for a smaller percentage of weather-related outages, they account for most of the **longest-lasting** outages through 2022.

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demand for cooling, which can overload the system. A recent study shows that widespread extreme heat has coincided with power outages in every U.S. region from 2012 to 2021. These events can expose people to dangerous temperatures and amplify health risks during heat waves.

Wildfire accounted for 39 outages (about 2% of weather-related outages). More than half of these outages were concentrated in the last five years. About one-third were public safety power shutoffs by utilities due to wildfires or to reduce risk of equipment-related ignitions during extreme fire weather days. Wildfire seasons are lengthening and intensifying across the U.S.

Outages by region

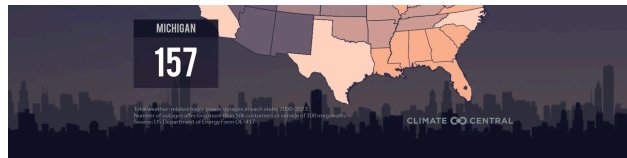
Multiple states can be impacted by a single outage because extreme weather events often affect large areas and the electrical grid is interconnected across state lines. Outages that affected multiple states and regions were counted in each state and region's total number of events, but were counted only once

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The number of weather-related outages varies among U.S. regions — reflecting the weather each region experiences, as well as relative population density and grid vulnerability. The Southeast (360), South (352), Northeast (350), and Ohio Valley (301) experienced the most weather-related outages from 2000 to 2023.

U.S. Climate Region	Weather-related outages (2000-2023)
Southeast	360
South	352
Northeast	350
Ohio Valley	301
Upper Midwest	205
West	151
Northwest	73
Southwest	28
Northern Rockies and Plains	24

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Click the downloadable graphic: [Weather Outages by State](#)

Outages by state

The states with the most reported weather-related power outages from 2000 to 2023 were: Texas (210); Michigan (157); California (145); North Carolina (111); and Ohio (88).

State	Weather-related outages (2000-2023)
Texas	210
Michigan	157
California	145
North Carolina	111
Ohio	88
Louisiana	85
Virginia	83
Georgia	83
Pennsylvania	82
Florida	77
Alabama	76

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refrigeration, heating, and air conditioning can be [dangerous or even deadly](#), especially during extended outages. Outages can also lead to [a range of potential health consequences](#), particularly for those who rely on electricity for critical medical equipment.

[Older people](#) and individuals with disabilities or certain [health conditions](#) may be especially vulnerable during weather-related power outages. Consequences can be compounded by the precipitating events, such as flooding, wildfire, or extreme temperatures.

Longer outages amplify health risks for vulnerable populations. A [2023 study](#) found that long-duration outages were most prevalent across the Northeast, South, and Appalachia from 2018 to 2020. [Arkansas, Louisiana, and Michigan](#) in particular experienced significantly more long-duration outages in counties with large socially- and medically-vulnerable populations.

Building a more resilient grid

Upgrading the nation's electrical infrastructure to become more

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the rate of warming, which can ease mounting stress on our power grid and allow more time to adapt our systems to a changing climate.

In the near-term, there are [promising and innovative solutions](#) to build electricity security now.

[Microgrids](#) are self-sufficient energy systems with a smaller, distinct geographic footprint, such as a [college campus](#), hospital complex, or neighborhood. Their relatively small scale also makes microgrids more easily powered by renewable energy sources like [solar and wind power](#), which has the added benefit of reducing emissions from power generation.

[Smart grid technologies](#) include sensors that allow operators to assess grid stability and provide consumers with better information about outages.

[Hardening the grid](#) refers to measures that fortify the system against damage. This can include tree trimming along power lines, replacing wooden electrical poles with steel or

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encourage customers to cut back on usage during peak times.

LOCAL STORY ANGLES

Weather and climate disasters in your state or region

Use NOAA's [Disaster Mapping tool](#) to visualize the frequency and cost of billion-dollar weather and climate disasters at national, regional, and state levels. The U.S. Energy Information Administration's [Energy Disruptions Maps](#) help identify local risks to energy infrastructure from tropical storms, wildfire, and flooding. Local utility companies often provide current power outage maps and updates by zip code.

Tools for reporting on extreme weather events and disasters

Journalism schools and organizations provide advice for [responsibly reporting on disasters](#), including focusing on [safety](#), [data](#), and [cultural sensitivity](#). Additional resources include Climate Central's [extreme weather toolkits](#) (in English and Spanish).

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Outages:

Preparedness and safety guidance for power outages are available from the [Federal Emergency Management Agency](#) and the [Centers for Disease Control and Prevention](#).

CONTACT EXPERTS

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Assistant Professor
University of Washington

Relevant expertise: Environmental health, vulnerable populations during power outages

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FIND EXPERTS

Submit a request to [SciLine](#) from the American Association for the Advancement of Science or to the [Climate Data Concierge](#) from Columbia University. These free

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[services](#) at regional NOAA, USDA, and Department of the Interior offices.

Explore databases such as [500 Women Scientists](#), [BIPOC Climate and Energy Justice PhDs](#), and [Diverse Sources](#) to find and amplify diverse expert voices.

Reach out to your [State Climate Office](#) or the nearest [Land-Grant University](#) to connect with scientists, educators, and extension staff in your local area.

METHODOLOGY

Power outage data from 2000 to 2023 were collected from the U.S. Department of Energy’s [Form OE-417](#) reports. Major outages are events that affect at least 50,000 customers (homes or business) or interrupt service of 300 megawatts or more. For the purpose of our analysis, we consider only power outages (including blackouts and voltage losses), fuel supply emergencies, and emergency appeals for reduced electricity usages where there was a reported number of customers affected or power lost — and where outages were attributed to weather- or wildfire-related causes. We do not

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refer to a broad range of conditions. Notably, these conditions may not meet meteorological criteria for [severe weather](#). Climate Central analysts reviewed report details and relevant media coverage to appropriately assign documented weather events to individual outages. Power outages that affected multiple states and regions were counted in each state and region's total number of events, but were counted only once in the national number of events. Regions were defined based on [NOAA's U.S. Climate Regions](#).

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